

PRACTICAL WEEK1:

Task1:

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user:

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both the parent and child processes.

```
UW PICO 5.09 File: task1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char input[100];
    char *args[20];
    int i = 0;

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    // Split command into arguments
    char *token = strtok(input, " ");

    while (token != NULL && i < 19) {
        args[i] = token;
        i++;
        token = strtok(NULL, " ");
    }

    args[i] = NULL;

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        // CHILD
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execvp(args[0], args);

        // Only executes if execvp fails
        perror("exec failed");
        exit(1);
    }
    else {
        // PARENT
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("\nChild process finished.\n");
    }

    return 0;
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

```

vishalkarnati@VISHALs-MacBook-Air ~ % cd ~/Desktop
vishalkarnati@VISHALs-MacBook-Air Desktop % cd ~/Desktop/os
vishalkarnati@VISHALs-MacBook-Air os % mkdir week1pract
vishalkarnati@VISHALs-MacBook-Air os % cd week1pract
vishalkarnati@VISHALs-MacBook-Air week1pract % nano task1.c
vishalkarnati@VISHALs-MacBook-Air week1pract % gcc task1.c -o task1
vishalkarnati@VISHALs-MacBook-Air week1pract % ./task1
Enter a command: ls -l

Parent Process
Parent PID: 6875
Child PID: 6879

Child Process
Child PID: 6879
Parent PID: 6875
total 80
-rwxr-xr-x  1 vishalkarnati  staff   34024 17 Aug 19:11 task1
-rw-r--r--  1 vishalkarnati  staff    1162 17 Aug 19:11 task1.c

Child process finished.
vishalkarnati@VISHALs-MacBook-Air week1pract % pwd
/Users/vishalkarnati/Desktop/os/week1pract
vishalkarnati@VISHALs-MacBook-Air week1pract % date
Mon Aug 17 19:12:22 IST 2026
vishalkarnati@VISHALs-MacBook-Air week1pract % ls -a
.  ..  task1  task1.c
vishalkarnati@VISHALs-MacBook-Air week1pract % ls -l
total 80
-rwxr-xr-x  1 vishalkarnati  staff   34024 17 Aug 19:11 task1
-rw-r--r--  1 vishalkarnati  staff    1162 17 Aug 19:11 task1.c
vishalkarnati@VISHALs-MacBook-Air week1pract % █

```

Task2:

Using Linux terminal commands ([uname](#), [lscpu](#), [lsblk](#), [ps](#), [top](#)), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
[vishalkarnati@VISHALS-MacBook-Air week1pract ~ % uname -a
Darwin VISHALS-MacBook-Air.local 25.2.0 Darwin Kernel Version 25.2.0: Tue Nov 18 21:09:34 PST 2025; root:xnu-12377.61.12~1/RELEASE_ARM64_T8112 arm64
[vishalkarnati@VISHALS-MacBook-Air week1pract ~ % system_profiler SPHardwareDataType
Hardware:

Hardware Overview:

  Model Name: MacBook Air
  Model Identifier: Mac14,2
  Model Number: MLY43LL/A
  Chip: Apple M2
  Total Number of Cores: 8 (4 performance and 4 efficiency)
  Memory: 8 GB
  System Firmware Version: 13822.61.10
  OS Loader Version: 13822.61.10
  Serial Number (system): MAXILLV26L
  Hardware UUID: E10886E4-50DB-5CB6-B9C3-48B7CCCF6C01
  Provisioning UDID: 00080112-000478641A3401E
  Activation Lock Status: Disabled

[vishalkarnati@VISHALS-MacBook-Air week1pract ~ % sysctl -n hw.ncpu
8
[vishalkarnati@VISHALS-MacBook-Air week1pract ~ % vm_stat
Nach Virtual Memory Statistics: (page size of 16384 bytes)
Pages free: 6044.
Pages active: 97380.
Pages inactive: 98989.
Pages speculative: 561.
Pages throttled: 0.
Pages wired down: 98768.
Pages purgeable: 2896.
"Translation faults": 46472000.
Pages copy-on-write: 3611348.
Pages zero filled: 17226873.
Pages reactivated: 7228069.
Pages purged: 1301904.
File-backed pages: 69756.
Anonymous pages: 124894.
Pages stored in compressor: 479879.
Pages occupied by compressor: 196743.
Decompressions: 5496185.
Compressions: 7781180.
Pageins: 2776274.
Pageouts: 31044.
Swapsins: 395.
Swapsouts: 11724.

[vishalkarnati@VISHALS-MacBook-Air week1pract ~ % diskutil list
/dev/disk0 (internal, physical):
#0:                                     TYPE NAME                       SIZE      IDENTIFIER
0:      GUID_partition_scheme                                *600.3 GB   disk0
1:          Apple_APFS_Isc Container disk1                     524.3 MB   disk0s1
2:          Apple_APFS Container disk3                          494.4 GB   disk0s2
3:          Apple_APFS_Recovery Container disk2                 5.4 GB     disk0s3

/dev/disk3 (synthesized):
#0:                                     TYPE NAME                       SIZE      IDENTIFIER
0:      APFS Container Scheme                                *494.4 GB   disk3
1:          APFS Volume Macintosh HD                          12.3 GB     disk3s1
2:          APFS Snapshot com.apple.os.update-...             12.3 GB     disk3s1s1
3:          APFS Volume Preboot                               9.9 GB      disk3s2
4:          APFS Volume Recovery                             2.5 GB      disk3s3
5:          APFS Volume Data                                 133.2 GB    disk3s5
6:          APFS Volume VM                                   1.1 GB      disk3s6

#0:          APFS Volume data                                 133.2 GB    disk3s5
#6:          APFS Volume VM                                   1.1 GB      disk3s6

[vishalkarnati@VISHALS-MacBook-Air week1pract ~ % ps aux
USER          PID %CPU %MEM    VSZ   STAT  STARTED      TIME COMMAND
_vindowservr  390  8.3  0.7 436582528  62512  ??  R   10:52AM  48:20.81 /System/Library/PrivateFrameworks/SkyLight.framework/Resources/WindowServer -daemon
vishalkarnati 720  7.6  0.4 436417456  31168  ??  S   10:53AM  3:18.83 /usr/libexec/duetexpertd
root          7086  6.3  0.1 436345088  11472  ??  Ss  7:21PM  0:00.07 /usr/libexec/mdmclient daemon
vishalkarnati 6702  5.6  5.1 194942624  427632  ??  R   7:05PM  3:26.11 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Googl
e
vishalkarnati 592  3.7  0.2 436342864  12896  ??  S   10:52AM  0:22.47 /System/Library/PrivateFrameworks/BioStreams.framework/Support/BioAgent
vishalkarnati 7096  3.1  0.1 486673072  11488  ??  S   7:21PM  0:00.01 /Users/vishalkarnati/Library/Application Support/Google/GoogleUpdater/152.0.7933.0/GoogleUpdater.app/Contents/MacOS/GoogleUp
root          7821  2.5  0.2 436244320  13616  ??  S   0:16AM  1:00.26 /sbin/launchd
vishalkarnati 7087  3.1  0.1 486778400  12208  ??  S   7:21PM  0:00.02 /Users/vishalkarnati/Library/Application Support/Google/GoogleUpdater/152.0.7933.0/GoogleUpdater.app/Contents/MacOS/GoogleUp
vishalkarnati 7094  3.0  0.1 485704960  12192  ??  Ss  7:21PM  0:00.01 GoogleUpdater --server --service=update-internal
root          367  2.8  0.1 436420272  12176  ??  Ss  10:52AM  0:12.12 /System/Library/PrivateFrameworks/ApplePushService.framework/apsd
vishalkarnati 7081  2.5  0.2 436346464  15872  ??  S   10:52AM  0:00.04 GoogleUpdater --server --service=update
root          7805  2.5  0.1 436348944  9008  ??  S   7:21PM  0:00.05 /System/Library/CoreServices/ManagedClient.app/Contents/MacOS/ManagedClient
vishalkarnati 6743  2.4  0.9 436598912  78944  ??  S   7:06PM  0:15.25 /System/Applications/Utilities/Terminal.app/Contents/MacOS/Terminal
vishalkarnati 7090  2.3  0.1 486779280  12496  ??  S   7:21PM  0:00.01 /Users/vishalkarnati/Library/Application Support/Google/GoogleUpdater/152.0.7933.0/GoogleUpdater.app/Contents/MacOS/GoogleUp
locationd     373  2.0  0.2 436345872  12928  ??  Ss  10:52AM  0:37.73 /usr/libexec/locationd
root          365  2.0  0.2 436345872  12928  ??  Ss  10:52AM  0:30.53 /usr/libexec/opendirectoryd
vishalkarnati 7089  1.9  0.1 486738688  11568  ??  S   7:21PM  0:00.01 /Users/vishalkarnati/Library/Application Support/Google/GoogleUpdater/152.0.7933.0/GoogleUpdater.app/Contents/MacOS/GoogleUp
vishalkarnati 7092  1.8  0.1 486737968  11536  ??  S   7:21PM  0:00.01 /Users/vishalkarnati/Library/Application Support/Google/GoogleUpdater/152.0.7933.0/GoogleUpdater.app/Contents/MacOS/GoogleUp
vishalkarnati 1083  1.6  3.0 537416800  240880  ??  S   10:10AM  7:32.39 /Applications/Google Chrome.app/Contents/MacOS/Google Chrome
root          466  1.5  0.1 436341952  12512  ??  Ss  10:52AM  0:01.22 /System/Library/Frameworks/Security.framework/Versions/A/XPCServices/authd.xpc/Contents/MacOS/authd
root          325  1.4  0.2 436421184  15776  ??  Ss  10:52AM  0:39.37 /usr/libexec/iodd
vishalkarnati 589  1.3  0.3 436372928  22288  ??  S   10:52AM  1:05.21 /System/Library/CoreServices/WindowManager.app/Contents/MacOS/WindowManager
vishalkarnati 567  1.2  0.1 436310224  6544  ??  S   10:52AM  0:17.01 /usr/sbin/cfprestd agent
root          545  1.0  0.4 436305344  36448  ??  Ss  10:52AM  0:52.96 /System/Library/DriverExtensions/com.apple.DriverKit-AppleBCMMLAN.dext/com.apple.DriverKit-AppleBCMMLAN.com.apple.bcmmlan b
root          391  1.0  0.0 436305120  3232  ??  Ss  10:52AM  0:12.54 /usr/sbin/notifd
vishalkarnati 713  1.0  0.2 436557696  20384  ??  S   10:53AM  0:20.63 /System/Library/PrivateFrameworks/TextInputUIHelper.framework/Versions/A/XPCServices/CursorUIViewService.xpc/Contents/Mac
vishalkarnati 715  0.9  0.6 436084368  52048  ??  S   10:53AM  0:31.26 /System/Library/CoreServices/Spotlight.app/Contents/MacOS/Spotlight
vishalkarnati 7083  0.9  0.1 486672672  11488  ??  S   7:21PM  0:00.01 /Users/vishalkarnati/Library/Application Support/Google/GoogleUpdater/152.0.7933.0/GoogleUpdater.app/Contents/MacOS/GoogleUp
_mdnsresponder 462  0.8  0.1 436354208  9104  ??  Ss  10:52AM  0:12.38 /usr/sbin/mdnsResponder
root          400  0.8  0.1 436306464  4480  ??  Ss  10:52AM  0:13.29 /usr/sbin/cfprestd daemon
vishalkarnati 1236  0.7  0.7 1949396384  60832  ??  S   11:09AM  4:40.28 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Googl
e
vishalkarnati 594  0.7  0.1 436385952  11680  ??  S   0:03.03 /usr/libexec/linkd
vishalkarnati 1091  0.7  1.1 487022256  88480  ??  S   10:59AM  36:33.94 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Google
vishalkarnati 666  0.7  0.2 436363424  17200  ??  S   10:13AM  0:16.78 /System/Library/PrivateFrameworks/CoreDuetContext.framework/Resources/ContextStorageEnt
vishalkarnati 1092  0.6  0.9 486132432  71504  ??  S   10:59AM  0:58.25 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Google
vishalkarnati 6670  0.6  0.2 436349088  14128  ??  S   7:04PM  0:01.09 /usr/libexec/networkserviceproxy
_rootstd      406  0.6  0.1 436342240  9696  ??  Ss  10:52AM  0:35.45 /usr/libexec/trustd
root          14205  0.5  0.1 436339680  5080  ??  Ss  10:00PM  0:00.93 /usr/libexec/securityd_system
vishalkarnati 844  0.5  1.5 436308800  12032  ??  S   10:53AM  0:54.16 /Applications/WhatsApp.app/Contents/MacOS/WhatsApp
vishalkarnati 658  0.5  0.4 411930784  31808  ??  S   10:53AM  1:57.89 /opt/homebrew/opt/mongodb-community@7.0/bin/mongo --config /opt/homebrew/etc/mongod.conf
_networkd     458  0.4  0.1 436739632  11440  ??  Ss  10:52AM  0:17.12 /usr/libexec/symptomsd
root          453  0.4  0.2 436383740  18944  ??  Ss  10:52AM  2:25.31 /usr/libexec/sisrptd
root          405  0.4  0.1 436355648  12176  ??  Ss  10:52AM  1:14.16 /usr/libexec/runningboardd
vishalkarnati 1317  0.3  0.3 1951224928  27184  ??  S   11:13AM  0:52.28 /Applications/Visual Studio Code.app/Contents/Frameworks/Code Helper.app/Contents/MacOS/Code Helper --type=utility --utilit
y
root          329  0.3  0.1 436310624  4784  ??  Ss  10:52AM  0:20.63 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/FSEvents.framework/Versions/A/Support/fsentstd
root          404  0.2  0.0 436326064  3080  ??  Ss  10:52AM  0:00.57 /System/Library/CoreServices/coreservicesd
_mdms_stores  528  0.2  0.5 441918128  42896  ??  S   10:52AM  0:39.33 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mds_stores
vishalkarnati 1315  0.2  0.6 1951223712  47616  ??  S   11:13AM  1:25.27 /Applications/Visual Studio Code.app/Contents/Frameworks/Code Helper (Plugin).app/Contents/MacOS/Code Helper (Plugin) --typ
e
root          354  0.2  0.2 436391328  17264  ??  Ss  10:52AM  0:44.03 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks/Metadata.framework/Versions/A/Support/mds
vishalkarnati 1093  0.1  0.3 486084000  25200  ??  S   10:59AM  0:06.92 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Google
_analyticsd   408  0.1  0.1 436349696  18576  ??  Ss  10:52AM  0:03.83 /System/Library/PrivateFrameworks/CoreAnalytics.framework/Support/analyticsd
root          510  0.1  0.1 436348608  9200  ??  Ss  10:52AM  0:29.01 /usr/libexec/symptd
root          368  0.1  0.1 436342032  9232  ??  Ss  10:52AM  0:06.44 /System/Library/CoreServices/launchservicesd
root          393  0.1  0.1 436338864  6896  ??  Ss  10:52AM  1:31.22 /usr/libexec/corebrightnessd --launchd
vishalkarnati 591  0.1  0.2 436359488  14464  ??  S   10:52AM  0:05.42 /System/Library/PrivateFrameworks/AssistantServices.framework/Versions/A/Support/assistantd
```

top: